Advanced Mathematical Formulations for Deterministic Video Overlay Extraction

The extraction of high-fidelity, compositing-ready overlays from compressed video streams, without the utilization of deep learning or external neural APIs, represents a profoundly ill-posed inverse problem in classical computer vision. The theoretical foundation of this process relies on the standard matting equation, which dictates that any observed image pixel is a linear combination of an unknown foreground color and a known (or estimated) background color, modulated by an unknown fractional opacity. Resolving this equation deterministically using exclusively standard computational libraries necessitates traversing highly complex topological and spatiotemporal domains.[1]

The historical evolution of extraction architectures within this domain demonstrates a clear trajectory from heuristic, feed-forward per-pixel classifications to massive, coupled spatiotemporal energy minimization frameworks, and ultimately to non-linear partial differential equation (PDE) solvers.[1] The objective of this comprehensive report is to exhaustively analyze the mathematical failures of legacy heuristic pipelines, dissect the experimental sandbox iterations of advection and reaction-diffusion models, and architect a unified, innovative framework that resolves the topological divergence between single-subject ("Island") and multi-hole overlay ("Donut") asset extraction.

Autopsy of Deterministic Heuristics and Variational Foundations

The foundational iterations of deterministic overlay extraction utilized a heavily optimized, heuristic-driven stack. This legacy pipeline initiated with frame-1 background sampling, relied on LAB color space Mahalanobis distance thresholding to isolate subjects, and employed morphological border cleanup to generate a distilled barrier matte. Temporal consensus algorithms stabilized the sequence, and a sequential despill pass attempted edge decontamination.[1] While computationally lightweight, this architecture fundamentally collapsed when processing standard compressed video.

The Physics of Texture-Boost Chatter

A primary failure modality in early heuristic pipelines emerged during "white-on-white" scenarios. When soft foreground details, such as animated glows, fine hair, or flower petals, exhibited chromaticity and luminance virtually identical to the white background, deterministic distance metrics failed to detect the boundary.[1] Early attempts to mathematically force separation relied on a pre-processing texture-boost experiment utilizing a Laplacian sharpening operator. If the observed continuous image is represented by the function $I(x, y)$, the classical unsharp mask operates by subtracting the scaled Laplacian of the image from the original signal to emphasize high-frequency edges:

$$I_{boost} = I - \kappa \nabla^2 I$$

Where κ represents a scalar controlling the boost intensity. In a theoretical, noise-free, infinite-resolution scenario, the Laplacian $\nabla^2 I$ evaluates to zero in flat background regions and produces large magnitude spikes precisely at object boundaries. However, compressed video formats, such as H.264 or High Efficiency Video Coding (HEVC) utilizing 4:2:0 chroma subsampling, are inherently contaminated by high-frequency macroblock noise and quantization errors. If this degradation is denoted as the noise field $\eta(x, y)$, the actual observed signal becomes $I_{true} + \eta$. Consequently, the boosted image equation expands to:

$$I_{boost} = I_{true} + \eta - \kappa \nabla^2 I_{true} - \kappa \nabla^2 \eta$$

The critical mathematical failure arises because, in flat white background regions, the true physical gradient $\nabla^2 I_{true}$ is zero, but the noise gradient $\nabla^2 \eta$ is highly non-zero.[1] The second derivative operator massively amplifies the magnitude of the compression artifacts. When this distorted I_{boost} signal is processed by the deterministic Mahalanobis distance metric $D_M(I_{boost}, \mu_B, \Sigma_B)$, the amplified noise structures force background pixels far outside the covariance threshold Σ_B. The algorithm falsely classifies this amplified background noise as complex foreground structure, resulting in highly visible "white chatter" and flickering halo artifacts.[1] Furthermore, temporal consensus heuristics incorrectly lock onto this chatter, attempting to stabilize the noise and thereby creating a persistent, vibrating halo around the subject.[1]

The Inadequacy of Sequential Edge Decontamination

The secondary failure modality of the heuristic stack involves Red-Green-Blue (RGB) edge contamination. The legacy pipeline handled color decontamination (despill) sequentially, operating under the assumption that if the alpha matte α and the background color B are known, the true foreground color F can be algebraically derived by inverting the compositing equation:

$$F = \frac{I - (1 - \alpha)B}{\alpha}$$

This sequential approach is mathematically unstable near soft edges where the alpha value is fractional (e.g., $0.1 < \alpha < 0.9$). Because both α and F are fundamentally unknown at these boundaries, calculating F relies entirely on the absolute precision of the estimated α. If the preceding heuristic pipeline underestimates α even fractionally, the denominator becomes artificially small, causing the estimated foreground color F to explode toward unnatural color spectra.[1] Conversely, if α is overestimated, F remains heavily polluted by the background color B. Because the variables are inextricably coupled, a sequential pipeline cannot recover from an early alpha estimation error, resulting in hard, contaminated borders.[1]

The Variational Alpha-Color-Confidence (VACC) Optimization Baseline

To transcend these severe limitations, the Variational Alpha-Color-Confidence (VACC) Optimization framework was developed. VACC represents a fundamental architectural departure from heuristic pipelines by framing the overlay extraction as a massive, coupled spatiotemporal energy minimization problem. Rather than solving for the alpha matte sequentially, VACC jointly optimizes the alpha matte α, the unmixed foreground color F, and a novel spatial confidence map W.[1] The global energy functional $\mathcal{E}(\alpha, F, W)$ is defined to find the minimum state across the video volume:

$$\arg\min_{\alpha,F,W} \mathcal{E}(\alpha, F, W) = \mathcal{E}_{data} + \lambda_{mat}\mathcal{E}_{mat} + \lambda_{col}\mathcal{E}_{col} + \lambda_{tv}\mathcal{E}_{tv} + \lambda_{edge}\mathcal{E}_{edge}$$

The VACC architecture introduces several highly original mechanisms to attack specific failure modalities. The Confidence-Gated Data Term ($\mathcal{E}_{data}$) multiplies the standard L_2 pixel residual by the latent confidence map W_i^t. By driving $W_i^t \to 0$ in ambiguous regions (such as white hair against a white background heavily corrupted by macroblock noise), the system is granted a mechanism to selectively ignore the input data. This effectively turns off the data constraint locally, allowing the values of α and F to be governed entirely by surrounding spatial and temporal smoothness priors.[1]

The Spatiotemporal Matting Laplacian ($\mathcal{E}_{mat}$) leverages a massive, multi-frame Laplacian regularization term, $\alpha^T(L_S + \gamma L_T)\alpha$. The matrix L_S represents the classical spatial affinity matrix, ensuring alignment with physical image edges, while the matrix L_T connects pixels along the time axis using dense optical flow arrays. Minimizing this quadratic form forces the alpha matte to transition smoothly through space and time, rendering temporal popping mathematically impossible.[1]

The Confidence Total Variation Prior ($\mathcal{E}_{tv}$) enforces an L_1 norm on the gradient of W, dictating a strictly piecewise-flat confidence map and mathematically forbidding high-frequency chatter.[1] Simultaneously, the Cross-Gradient Edge Penalty ($\mathcal{E}_{edge}$) eliminates the need for sequential despill by penalizing the inner product of the spatial gradients of the alpha matte and the foreground color: $|\langle \nabla \alpha, \nabla F \rangle|^2$. This mathematically forces the gradient of the foreground color to be orthogonal to the opacity boundary, preventing background bleeding by propagating the pure interior foreground color cleanly through the semi-transparent boundary.[1]

Because minimizing this global functional simultaneously is non-convex and computationally intractable for high-definition video resolutions, VACC relies on Alternating Minimization. By fixing any two variables, the energy functional becomes strictly convex with respect to the third. The system formulates each sub-problem as a highly sparse linear system and solves it utilizing Preconditioned Conjugate Gradient (PCG) descent and Iteratively Reweighted Least Squares (IRLS) for the non-differentiable L_1 Total Variation norm.[1] Operating within a narrow spatial band surrounding the unknown boundary reduces the size of the sparse matrices by over eighty percent, rendering the computation feasible on standard workstation hardware.[1]

The Empirical Sandbox: Defeating the Symmetrical Bias

While the VACC framework established a robust theoretical baseline, practical implementation required aggressive optimization to achieve real-time or near-real-time throughput for streaming applications. Subsequent empirical sandbox experiments isolated the temporal and spatial smoothing components to maximize efficiency. The evolution of the baseline model yielded significant insights into alpha mass distribution.

The "CNVM-HB" series of experiments focused on manipulating the confidence map and temporal Exponential Moving Average (EMA) to suppress chatter. The baseline metric state for full cleanup, Pass 03, demonstrated a mean alpha of 0.436570, border bleed of 0.006426, chatter of 0.020820, and temporal jitter of 0.027851. An aggressive anti-bleed iteration, Pass 11, successfully reduced border bleed to 0.006251 and temporal jitter to 0.027544, but lost critical anti-chatter gains, exhibiting a chatter rate of 0.019474. The definitive hybrid, Pass 15 (CNVM-HB-Lite v2), integrated advanced temporal EMA and spatial confidence weighting. This configuration preserved and drastically improved the anti-chatter win (0.002371) and temporal jitter (0.026415), while maintaining a mean alpha of 0.442476. However, it failed to sufficiently suppress border bleed, which remained elevated at 0.007850.

The mathematical autopsy of Pass 15 revealed a critical structural defect: the transition-zone alpha mass was geometrically "fat." The adaptive temporal EMA and confidence map smoothing were side-agnostic; they smoothed alpha values equally toward the interior (foreground) and exterior (background) sides of the boundary contour. If a pixel on the exterior side briefly received a non-zero alpha value due to compression artifacts or soft-edge overshoot, the temporal EMA locked that non-zero value into the running average rather than permitting it to decay to zero. The result was a puffy, dilated transition zone extending into the background.

The Failure of the Normal-Projected Exterior Sink (NPES)

Initial hypotheses assumed the residual bleed constituted a thin, detached exterior halo tail. To address this, the Normal-Projected Exterior Sink (NPES) was engineered. NPES utilized the boundary-normal projection to classify interior versus exterior pixels and applied a monotone sink force exclusively to the exterior side. The normal-projected exterior coordinate was defined as $e(p) = \max(0, d(p))$, where $d(p)$ is the signed distance from the $\alpha_0 = 0.5$ boundary. A Gaussian-complement sink potential $S(p) = 1 - \exp(-e(p)^2/2\sigma_d^2)$ was intended to map exterior pixels toward zero, gated by background color similarity and confidence floors.[1]

Empirical testing proved NPES to be an effective no-op. The residual error was not a detached exterior shell tail; the excess mass was inextricably baked into the continuous cross-boundary profile. NPES only shaved the extreme exterior pixels after the mass had already been diffused inward by the EMA and isotropic smoothing steps. The failure of NPES confirmed that the defect resided in the global transition-zone mass distribution, necessitating a fundamental redistribution of alpha mass across the entire transition gradient.

Shell Transport and Ray-Casting Interventions

To actively reshape the transition zone, the Paired Shell Transport methodology was developed. This algorithm quantified the transition zone into discrete integer shells based on the signed distance field. It explicitly forced mass from every exterior shell straight into its symmetric interior partner, operating under the assumption that moving mass inward would thin the transition zone without discarding total alpha mass. While directionally logical, empirical results proved disastrous. The target pass yielded a mean alpha of 0.443239, but border bleed escalated to 0.007907, chatter degraded to 0.006833, and temporal jitter worsened to 0.028199. The coarse, shell-quantized mass movements destabilized the profile shape, introducing spatial discontinuities that destroyed the temporal consensus.

Learning from the quantization failures of shell transport, the Normal Ray Profile Power Warp (NRPPW) model transitioned to continuous, sub-pixel ray casting. For every pixel in the narrow band, NRPPW sampled a one-dimensional ray along the local outward normal $n = \text{normalize}(\nabla \alpha_0)$. Utilizing bilinear interpolation, the model sampled the ray at continuous intervals and applied a monotonic, per-ray power warp:

$$\alpha'(t) = \alpha_{\text{ray}}(t)^{1+\beta \cdot \max(\phi_{\text{ray}}(t), 0)}$$

This formulation applied an exponent greater than 1.0 strictly on the exterior side ($\phi > 0$), forcing an exponentially faster decay precisely where the bleed originated. To prevent aggressive foreground thinning, the ray was explicitly renormalized to preserve local mass:

$\alpha_{\text{new}}(t) = \alpha'(t) \cdot (\sum \alpha_{\text{ray}} / \sum \alpha')$. NRPPW successfully warped the continuous profile at the source along the exact normal path, ensuring the bleed slope never formed, without introducing the discrete quantization jumps that doomed the Paired Shell Transport model.

Advanced Advection Models: The Original Spatial Advection (OSA) Family

To systematically contract the outer transition band, the Original Spatial Advection (OSA) family of algorithms treated the alpha matte as a fluid density field and advected it inward along the smoothed signed-distance normals. The core philosophy of OSA was to apply a continuous velocity field to pull the diffused exterior mass back toward the anchor core.

The Positive Divergence Defect in Mach 1 Advection

The baseline OSA implementation (Pass 51, designated "Mach 1") defined the velocity field as:

$$v = \kappa \cdot (\text{pivot} - \alpha) \cdot \text{envelope}$$

Where κ represents local curvature and the envelope restricts the velocity to the exterior band. This formulation successfully advected mass toward the core, significantly reducing the outer band mass proxy from 0.161912 (baseline) to 0.145596. However, the semi-Lagrangian remap operator possessed a critical mathematical flaw: positive divergence in the mid-transition zone. Because the envelope was isotropic and lacked a strict mass-conservation correction, the displaced mass diffused outward from the advection front. Consequently, while the outer tail shrank, the total transition band mass increased from 0.469416 to 0.479506, causing a global broadening of the ramp. Furthermore, applying uniform-strength velocity to moving soft edges ignored flow-warped phase mismatches, causing the temporal lag support metric to regress from 0.269082 to 0.281192.

Ablation Pass	Configuration Focus	Outer Band Mass	Transition Band Mass	Temporal Lag Support
Pass 15	Baseline v2 Conservative	0.161912	0.469416	0.269082
Pass 51	OSA Mach 1	0.145596	0.479506	0.281192
Pass 55	OSA v2 Corrected	0.149616	0.474591	0.276825
Pass 59	OSA Temporal-Only	0.150807	0.473705	0.275812

OSA-v2: Asymmetric Divergence-Free Vacuum Advection

To correct the ramp broadening, OSA-v2 attempted to enforce a divergence-free projection. The velocity field was modulated by a temporal lag map to prevent phase shifting on moving edges:

$$v = \kappa \cdot (\text{pivot} - \alpha) \cdot \text{envelope} \cdot (1 - \tau \cdot \text{lag\_map})$$

A finite-difference divergence correction was proposed: $v_{\text{corrected}} = v - \lambda \cdot (\nabla \cdot v) \cdot \mathbf{n} \cdot \max(\phi, 0)$. However, rigorous mathematical review demonstrated that this formulation is not a true Helmholtz-Hodge decomposition. Subtracting the scalar divergence multiplied by the normal vector does not yield a divergence-free field; rather, the $\max(\phi, 0)$ gate creates a Heaviside step discontinuity at the boundary, inducing high-frequency spatial oscillations.[3]

A mathematically sound correction required a confined Jacobi-iterated Poisson pressure solve. By computing the divergence $\text{div} = \nabla \cdot v$ and solving $\nabla^2 p = \text{div}$ strictly within the narrow band, the true corrected velocity is obtained via $v_{\text{corrected}} = v - \nabla p$. Despite implementing this rigorous divergence-free projection, empirical data from Pass 55 and Pass 59 demonstrated that advection along SDF normals contains an irreducible numerical diffusion component during the cv2.remap bilinear interpolation phase.[1] Even with perfect velocity fields, the act of remapping a discrete grid inherently diffuses the signal, proving that spatial advection is fundamentally incapable of simultaneously compressing outer mass and preserving strict transition widths.

The Thermodynamic Shift: Bistable Alpha Reactor (BAR)

The insurmountable numerical diffusion of advection models forced a paradigm shift from fluid transport mechanics to thermodynamic phase separation. The Bistable Alpha Reactor (BAR) abandons velocity fields entirely, treating the alpha matte as a chemical concentration and evolving it via a modified Allen-Cahn reaction-diffusion partial differential equation (PDE).[4]

The standard Allen-Cahn equation is the canonical PDE for producing sharp, stable interfaces from diffuse initial conditions. Its steady-state solution is a hyperbolic tangent sigmoid—the exact mathematically ideal profile for an alpha transition zone.[8] By running a limited number of micro-iterations, BAR pulls the mushy transition zone toward this optimal sigmoid shape without any spatial remapping, thereby generating zero numerical diffusion.[9]

Let $u = 2\alpha - 1$ represent the bistable variable, where $u = 1$ denotes the absolute foreground and $u = -1$ denotes the absolute background. The discretized local reaction for BAR-lite within the narrow band is defined as:

$$u_{\text{new}} = u + \Delta t \cdot \left(\epsilon \nabla^2 u + u(1 - u^2) \cdot (1 + \beta\phi) \cdot C^\gamma\right)$$

This formulation introduces highly specific modifications to prevent the PDE from indiscriminately destroying legitimate foreground geometry:

1. **Tilted Asymmetric Double-Well Potential:** The reaction term $u(1-u^2)$ is modulated by the geometric tilt $(1+\beta\phi)$. On the exterior side ($\phi > 0$), this tilt makes the background well ($\alpha = 0$) thermodynamically deeper than the foreground well. Alpha values slightly above 0.5 on the exterior are violently pulled toward zero, actively shrinking the transition zone from the outside.[5]
2. **Confidence-Gated Time Step:** The reaction is scaled by the existing confidence map C^γ. In ambiguous, low-confidence zones (such as white-on-white noise), the effective time step drops to near zero. The PDE halts, preserving the delicate anti-chatter smoothing established by the temporal EMA and preventing the nucleation of spurious Turing-like patterns.[13]
3. **Anisotropic Diffusion Suppression:** To prevent the Laplacian term $\epsilon\nabla^2 u$ from eroding legitimate soft foreground details like hair or motion blur, the diffusion coefficient is actively suppressed on the interior side ($\phi < 0$).

Ablation Pass	Configuration Focus	β (Tilt)	γ (Conf)	Δt	Expected Sandbox Outcome
Pass 60	Target BAR-lite	0.28	2.4	0.35	Profile sharpens, outer mass drops
Pass 61	Safer Regime	0.14	2.4	0.35	Minimal profile alteration
Pass 62	Strong Tilt	0.42	2.4	0.35	Extreme boundary steepening
Pass 63	Weak Confidence	0.28	1.2	0.35	High risk of chatter relapse
Pass 64	Failure Ceiling	0.28	2.4	0.55	Instability and foldover expected

Because BAR-lite executes via an in-place Jacobi step, it drives mid-band alpha values toward their nearest stable state, steepening the profile without spatial displacement. This decisively cures the transition-band mass broadening that plagued the OSA family.[15]

The "White Halo" Contamination: Luma-Aware Vortex and Resonance Models

While BAR-lite optimally resolves the geometric shape of the alpha transition profile, empirical analysis utilizing high-contrast testing assets (e.g., wispy blonde hair against a pure green chroma key or solid white background) indicated that a faint bright rim persisted on the highest-luma strands. This "white halo" is not a spatial geometry problem, but a severe color-contamination artifact originating from the pre-multiplication inherent in compressed video.

In scenes with bright backgrounds, pixels near the boundary with low fractional alpha ($\alpha \approx 0.1 - 0.3$) possess RGB values virtually identical to the background. Deterministic systems assign non-zero alpha to these pixels because the LAB Mahalanobis distance detects compression noise, and the temporal EMA locks that alpha into the running average. When this alpha plate is composited in software like OBS over a dark scene, the bright edge pixels multiply, glowing brilliantly white.[1]

Because previous models (NRPPW, OSA, BAR) were color-agnostic, they failed to recognize that high-luma transition mass is inherently more destructive to the composite than low-luma transition mass.

Ghost Repulsion and Plasma Collapse

The initial theoretical approach to this specific luma-inflation utilized extreme geometric mirroring. The Ghost Repulsion Plasma Collapse (GRPC) model hypothesized that outer transition mass behaves like a self-repelling "plasma cloud." To counteract this, GRPC generated a virtual negative-ghost field from the interior of the subject, mirrored precisely along the normal ray.

For a point p in the narrow band, a ray was sampled at continuous intervals $t \in \{-4, \ldots, 4\}$. The interior ghost was defined as $\text{ghost}(t) = \alpha_{\text{ray}}(-t)$ for $t > 0$.

The repulsion force was explicitly amplified by local curvature $\kappa = \nabla \cdot \mathbf{n}$, ensuring that tight curls of hair received massive restorative forces:

$$\text{repulsion}(t) = \gamma \cdot \text{ghost}(t) \cdot (1 + \lambda|\kappa|) \cdot \frac{\max(\phi_{\text{ray}}(t), 0)}{|\phi_{\text{ray}}(t)| + 1}$$

While GRPC was geometrically innovative, it remained color-blind. The critical breakthrough required marrying the ghost vortex mechanics with explicit luma gating.

Luma Premul Vortex Guard (LMPVG) and Polarity Adaptation

The Luma Premul Vortex Guard (LMPVG) constrained the ghost repulsion strictly to high-luma pixels. By sampling the luma along the ray, $Y_{\text{ray}} = 0.299R + 0.587G + 0.114B$, a gate was constructed to isolate bright zones: $\text{luma\_gate} = \text{clip}((Y_{\text{ray}} - 0.68)/0.22, 0, 1)$. The vortex force only activated on the exact pixels that would glow white in an OBS composite. However, resolving the white halo on bright backgrounds inadvertently caused structure loss and foreground thinning when the asset was placed against a black background. The ultimate resolution, the Polarity Adaptive Vortex Guard (PAVG), utilized the known background luma

(derived from frame-1 sampling) to dynamically switch the algorithm's behavior.
By calculating the delta between the ray luma and the background luma, $\Delta Y(t) = Y_{\text{ray}}(t) - Y_{bg}$, PAVG deployed full vortex strength exclusively when the pixel was brighter than the background (the white-halo case). When the pixel was darker than the background (the structure-loss case), the vortex was weakened by 60%, and the minimum alpha floor was dynamically boosted by $0.09 \cdot \alpha_0$. This polarity-aware adaptation achieved perfect, artifact-free composites across all background spectrums without requiring manual user intervention.

Luma-Weighted Inverse-Premultiply Clamp (LIPC)

Running parallel to the vortex models, the Luma-Weighted Inverse-Premultiply Clamp (LIPC) attacked color contamination using the inverse compositing equation. By deriving the theoretical unmixed foreground color $\hat{F} = \frac{I-(1-\alpha)B}{\alpha}$, LIPC operates as a highly sensitive, mathematically rigorous halo detector.[17]

If the alpha matte has been overestimated, dividing by α forces $\hat{F}$ to diverge wildly toward the background color. LIPC calculates the Euclidean color distance $\Delta_F = \|\hat{F} - B\|_2$. A localized halo score $h(p)$ is computed, peaking exclusively when the inversely-premultiplied foreground is simultaneously suspiciously bright and suspiciously identical to the background.[19]

$$h(p) = \text{clamp}\left(\frac{T_\Delta - \Delta_F(p)}{T_\Delta}, 0, 1\right) \cdot \text{clamp}\left(\frac{Y_F(p) - T_Y}{1 - T_Y}, 0, 1\right)$$

LIPC then applies a multiplicative reduction to the alpha matte guided by this score, $\alpha'(p) = \alpha(p) \cdot (1 - \lambda \cdot h(p) \cdot w(p) \cdot C(p))$, where $w(p)$ is a spatial Gaussian falloff. Rather than fighting the profile ramp width via physical advection or thermodynamic reshaping, LIPC selectively destroys alpha mass strictly where the inverse-compositing physics prove the mass is an illusion caused by background color bleeding.[21]

The Transcendent Fixes: Temporal Logit-Profile (TLP) and Shadow Resonance

As the pipeline matured, combining spatial stability with temporal coherence required operating in logarithmic odds space. The Temporal Logit-Profile (TLP) architecture bypassed direct alpha manipulation, mapping alpha into logit space: $z = \log(\alpha) - \log(1 - \alpha)$.
TLP performed a confidence-and-distance-weighted linear fit of logits against the signed distance field within the narrow band, yielding a continuous mathematical model of the boundary slope (B) and intercept (A). Crucially, TLP applied the temporal Exponential Moving

Average directly to these abstract fitted parameters (A and B) rather than to the per-pixel alpha values.
By tracking the geometric profile parameters across time, TLP constructed a temporally stable, idealized transition slope. The pixel-level logits were then pulled toward this idealized slope via a pivot-controlled, sigmoid-gated delta. This resulted in an astoundingly stable, smooth transition profile that eliminated high-frequency spatial noise and temporal jitter simultaneously. When combined with the Luma Edge Crush (TLEC) to form the TLP Polarity Luma Guard (TLPG), the architecture achieved an unprecedented balance of profile steepness and halo suppression.

Abyssal Echo Devourer (AED) and Shadow Twin Resonance (STR)

Within the experimental sandbox, the theoretical limits of alpha manipulation were tested using extreme, unconstrained models. The Abyssal Echo Devourer (AED) explicitly generated a "mirror universe" ghost from the time-reversed TLP state. For every exterior pixel, AED calculated an echo alpha on the exact opposite side of the zero contour. The real profile and its echo "devoured" each other, but only when local curvature signs matched, luma was excessively high, and temporal consistency with the flow warp was perfect.
Similarly, Shadow Twin Resonance (STR) allowed the real profile and its shadow twin to exchange mass via a resonance factor $\gamma \cdot \min(\alpha, \alpha_{\text{twin}}) \cdot (1 + \lambda|\kappa|)$. These models proved that dynamic, self-annihilating twin fields could theoretically obliterate the white halo instantaneously. However, their reliance on perfect curvature symmetry rendered them slightly too volatile for general production deployment compared to the robust simplicity of LIPC and BAR-lite.

Topological Bifurcation: The Overlay (CHHC) Method

A fundamental breakthrough in the Alpha Fix architecture was the realization that single-subject "Island" assets and multi-hole "Donut" overlays demand entirely different mathematical extraction paradigms. Subject extraction relies on refining a single continuous exterior boundary. Overlay extraction presents a global connectivity problem: because dark decorative frame pixels often match the central subject color distributions, heuristic distance metrics treat the entire 1920x1080 frame as foreground, completely blocking the underlying OBS display capture with 99.99% nonzero alpha.[1]
Attempting to resolve both topologies with a unified, topology-aware PDE requires immense computational overhead and risks destroying the clean subject case. Therefore, the architecture strictly bifurcates. Overlay topologies bypass the BAR-lite and LIPC refinements entirely, routing into the Contour-Hierarchy Hole Carving (CHHC) module.
CHHC utilizes a multi-seed topological flood methodology. It processes the raw thresholded alpha mask M_{raw} through a morphological closing operation to seal compression gaps, preventing internal holes from leaking into the background. It then extracts the topological contour hierarchy utilizing standard algorithms (e.g., OpenCV's RETR_CCOMP).[22] The

outermost contour lacking a parent node is defined as the rigid frame. All contour children of this frame are classified as potential internal holes (e.g., gameplay windows, chat boxes).[24]
To reject tiny decorative gaps in armor or texture noise, CHHC enforces strict size and position gating. An interior contour C_j is accepted as a transparent hole only if its total area exceeds a fractional threshold ($A_{\min}$) and its spatial centroid falls within a valid, centralized region $\mathcal{R}_{\text{valid}}$.[24]
The final overlay mask is constructed by boolean subtraction of the accepted child polygons from the raw frame mask. A localized Euclidean distance transform dictates a 2.5-pixel feathering radius exclusively on the inner edges, ensuring the frame remains crisp while the gameplay capture blends seamlessly without harsh aliasing.[27]

CHHC Ablation	Configuration Focus	Hole Area Threshold	Close Kernel	Target Metric Result
Pass 80	Baseline Overlay	N/A	N/A	99.99% Global Nonzero
Pass 81	Target CHHC	0.03	5px	~26.0% Nonzero, Open Center
Pass 83	Massive Frame Close	0.03	9px	Absolute protection against leak-through
Pass 84	Micro-Hole Acceptance	0.01	5px	Successful multi-chat window extraction

Conclusion

The evolution of deterministic video overlay extraction has comprehensively outgrown the mathematical limitations of sequential, per-pixel heuristics and semi-Lagrangian advection flows. Extensive empirical data explicitly proves that treating alpha matting as a spatial fluid transport problem inherently broadens transition profiles due to unavoidable numerical interpolation diffusion.
The finalized architecture of Alpha Fix resolves these barriers through strict topological bifurcation. For multi-hole overlay assets, the Contour-Hierarchy Hole Carving (CHHC) algorithm provides absolute topological certainty, automatically carving center and chat windows without geometry-agnostic leak-through. For complex subject assets, the integration of the Bistable Alpha Reactor (BAR-lite) and the Luma-Weighted Inverse-Premultiply Clamp (LIPC) creates an unparalleled refinement engine. By redefining the alpha matte as a bistable chemical concentration, the Allen-Cahn PDE naturally drives fuzzy, artifact-heavy transitions into mathematically pristine sigmoids without spatial diffusion. Simultaneously, the inverse-compositing color error dynamically targets and annihilates white halos without sacrificing legitimate soft edge detail. This unified approach fundamentally transcends the

capabilities of traditional chroma-key algorithms, delivering flawless, artifact-free composites on complex, compressed, and AI-generated video assets.

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